

Increasing Access by Waiving Tuition: Evidence from Haiti

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Despite recent gains in increasing access, an estimated 58 million children worldwide are still out of school. Abolishing school fees has increased enrollment rates in several countries where enrollments were low and fees were high. However, such policies may be less effective, or even have negative consequences, when supply-side responses are weak. This article evaluates the impacts of a tuition waiver program in Haiti, which provided public financing to nonpublic schools conditional on not charging tuition. We conclude that a school's participation in the program results in more students enrolled, more staff, and slightly higher student-teacher ratios. The program also reduces grade repetition and the share of overage students. While the increase in students does not directly equate to a reduction in the number of children out of school, it does demonstrate strong demand from families for the program and a correspondingly strong supply response from the nonpublic sector.

Introduction

Despite impressive gains over the last 20 years, the Millennium Development Goal (MDG) for 2015 of Universal Primary Education was missed by a wide margin. It is estimated that in 2011, 58 million primary school age children were still not enrolled. While more than half of these children (nearly 30 million) live in sub-Saharan Africa, there are nearly 3 million out-of-school children in the Latin America and Caribbean region (UNESCO 2015). About 200,000 of these children live in Haiti, where tuition fees charged by a predominantly nonpublic sector have historically presented a large barrier to families wishing to send children to school. However, progress in Haiti has been steady: by 2010, an estimated 90 percent of primary school aged children were enrolled in school, up from about 60 percent in 2001 (World Bank 2014a; UNESCO 2015).

Around the world, both the direct costs of going to school—tuition and other fees, uniforms, transport, books, and so forth—and the opportunity costs, particularly in the form of lost labor for the household, remain barriers to achieving universal primary enrollment and completion. A range of interventions aimed at reducing these costs have been rolled out across regions, falling into roughly four categories: cash transfers to households; vouchers

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to households to facilitate school choice; providing goods or services that are required for school for free (e.g., uniforms or books); and abolishing enrollment, tuition, and other fees. While these approaches to stimulating demand for schooling may be effective, they may not result in increased enrollment, attendance, and ultimately learning if the supply-side response is inadequate. An outright absence of schools could make demand-side interventions ineffective for the most poorly served communities, while overcrowded classrooms, excessively high student-teacher ratios, and lack of materials could deteriorate the learning environment for all students and deter new students from entering. The literature on the impacts of each type of intervention continues to grow, but broadly speaking, the results are positive: enrollment and grade completion increase (although impacts on learning are often zero; Fiszbein et al. 2009; Krishnaratne et al. 2013).

Research from several countries, particularly in Africa, shows that abolishing school fees over the last two decades has had positive impacts on enrollment in primary school (World Bank and UNICEF 2009). Gross enrollment rates are estimated to have increased by 73 percent in Uganda, 100 percent in Malawi, and 12 percent in Mozambique shortly after each of these countries declared primary school to be free (Bategeka and Okurut 2005; Fiszbein et al. 2009; Petrosino et al. 2012). In Kenya, Lucas and Mbiti (2012) exploit prepolicy geographic variation in dropout rates to estimate the impacts of the 2003 abolition of primary school fees, concluding that the program increased access and completion rates, particularly among poorer students. South Africa's experience was less successful, however, likely because of a context characterized by already high enrollment rates and relatively low school fees (Borkum 2009).

Relatedly, a small but growing literature on the effects of burgeoning private school sectors in many developing countries suggests that low-cost private provision has contributed to increased enrollments and that public financing of such providers may be an effective way to further increase access to education (Dixon 2012; Andrabi et al. 2015). In the program context most similar to the one evaluated in this article, Barrera-Orsorio and Raju (2011) find that enrollment subsidies to private schools in the Punjab province of Pakistan substantially increase enrollments and some schooling materials in participating schools.

In Haiti, over 90 percent of primary schools were nonpublic in the early 2000s, and the vast majority of these schools charged tuition. Coupled with the cost of mandatory uniforms, books, and other inputs, the direct costs of schooling were prohibitively high for many families. Estimates showed that total average costs (direct and indirect, including uniforms, transportation, and feeding) of sending a child to school were about \$131 per child per year (Merisier 2004; World Bank 2007). At a time when gross national income (GNI) per capita was estimated to be about \$400, nonpublic education was

essentially unaffordable. In addition, financial constraints were also believed to be a leading cause of the problem of overage students in school, as parents may have had to cycle children in and out of school, depending on their cash flow. The average age of students in grade 6, for instance, was 16 years old according to the 2003 school census, when the corresponding age for that grade should be 11 or 12 years old.

Given the importance of schooling costs as a barrier to access and the large role of the nonpublic sector, a demand-side response in the form of a program to abolish school tuition fees was initiated in 2007. This article utilizes the experimental nature of the program, whereby schools were randomly selected to participate from among qualified applicant schools, to estimate the impact of this program on school enrollments, student-teacher ratios, grade advancement, and other indicators at the school level.¹

We believe that the results of this evaluation are of interest beyond Haiti, as key features of the Haitian system—low state capacity and weak public delivery of education services—are common to many low-income countries, while the rapid growth of nonpublic schools in many of these countries makes the Haitian case, with a large and vibrant nonpublic sector, increasingly relevant. The next section of this article describes the Haitian context and the program itself; the following section describes the program randomization, data and estimation method, and results; and the final section presents conclusions.

An Approach to Increasing Access: Tuition Waivers

The Haitian Context: Many Nonpublic Schools and Out-of-School Children

Haiti's education system today stands out for its preponderance of private actors, the result of a rapid expansion of private provision throughout the latter part of the twentieth century (Easton and Fass 1989; Demombynes et al. 2010). Though Haiti was one of the first nations worldwide to state that education should be free and compulsory in the country's founding constitution of 1805 (Haggerty 1989), the actual delivery of education as a public service to the majority of the population failed to materialize. As Fass (1988) details, Haiti's educational system in the nineteenth and early twentieth centuries was characterized by periods of modest public expansion interspersed between long eras of government neglect. An accord between the Catholic Church and the government, as well as the growth of Protestant missions, contributed substantially to expanding school supply, and net primary school enrollment is estimated to have peaked at about 15 percent at the turn of the century. Following the turbulent period of US occupation, primary school

¹ While these indicators relate to education access and conditions within schools, the data available do not allow an examination of learning outcomes.

enrollment grew roughly 1–2 percent annually through the first half of the twentieth century, with the number of public and nonpublic schools growing at roughly similar rates (Fass 1988; Demombynes et al. 2010). With the onset of the Duvalier regimes in 1957, public expansion continued its slow pace, while nonpublic provision grew dramatically. Between the early 1960s and the early 2000s, the number of public primary schools in Haiti increased from about 800 to about 1,300. In contrast, the number of nonpublic primary schools grew from about 800 to nearly 14,000 during the same period (Demombynes et al. 2010). Consequently, in the early 2000s, public schools accounted for fewer than 10 percent of primary schools but served about 20 percent of all primary students because of their larger average size.

Nonpublic schools are a highly heterogeneous group. They include religious schools, community schools, schools started by nongovernmental organizations (NGOs), and for-profit schools. The impetus for their creation varies. A historical trend of religious groups motivated by proselytization and charity opening schools accelerated under the second Duvalier regime of “Baby Doc,” who instituted a rule that any newly opened church was required to also create a school. By the early 2000s, religiously affiliated schools, including affiliations with local and international churches and religious NGOs, accounted for about 45 percent of nonpublic primary schools. In other cases, the motivation was a desire to serve the less fortunate through secular NGOs, while others were at least partly driven by profit-seeking and established schools as for-profit enterprises. This diverse group of privately run secular schools accounted for about 40 percent of all nonpublic schools. Finally, for some the motivation was necessity, given the state’s limited ability to provide services, especially in rural areas. Under these conditions, community leaders often organized to respond to the latent parental demand for schooling. These community schools accounted for the remaining 15 percent of all nonpublic primary schools.²

Despite the rapid and diverse growth in the nonpublic school sector, by 2005, the primary net enrollment rate in Haiti was estimated to be as low as 50 percent, far below the sub-Saharan Africa average of 70 percent.³ This low rate was believed to be directly related to the costs of education: according to the 2002–3 Haitian School Census, over 90 percent of all nonpublic primary schools charged tuition, averaging about US\$20 per student per year (equivalent to 5 percent of GNI per capita) plus other fees (for enrollment, exams, and so forth). Consequently, large numbers of children were out of school or overage for their grade (e.g., in preschool rather than primary;

² For a more complete discussion of the evolution of the Haitian education system, see Demombynes et al. (2010).

³ The estimate from Haiti comes from Cayemittes et al. (2007), while the sub-Saharan Africa average comes from the World Bank’s Global Education Statistics, available at <http://databank.worldbank.org/data/home.aspx>.

Cayemittes et al. 2007). If the country was to make rapid progress toward the goal of universal primary education, Haiti's government and society needed to increase access, which meant, at least in the short term, working with the nonpublic sector.

Determinants of Educational Demand and the Tuition Waiver Program

In the classical economic formulation, education is considered an investment, with demand for education resulting from a rational weighing of the costs against the expected stream of future benefits (Becker 1964). The benefits, or returns to education, are primarily measured in terms of higher wages in the labor market and globally have been estimated to be substantially higher than average returns to alternative investments (Psacharopoulos and Patrinos 2010). Important nuances to this formulation have been developed in recent years, as some authors have pointed out that returns to education extend far beyond higher wages and can include better health and happiness, while others have made the case that the perceived returns to education, rather than actual economywide averages, actually drive decision making (Jensen 2010; Oreopoulos and Salvanes 2011).

For the poor in Haiti, evidence suggests that perceived returns to primary education have been quite high for several decades. As described in Easton and Fass (1989), Haitian parents value primary education as their children's best possible route out of poverty, as a prerequisite for further education and professional employment, or as a de facto minimum requirement for international migration.⁴ These beliefs have been reaffirmed by more recent qualitative research with parents in Haiti, who believe education is critical to realizing their children's full potential and for enabling their children to later help provide care in their old age (Jayaram 2015).

However, low economic growth and the many market distortions that contribute to poverty kept effective demand, and thus enrollment rates, low. With nearly one-third of Haitians living in extreme poverty in 2000 and the scarcity of public schools, paying for primary education did not make economic sense because of the extremely high marginal value of money, or was simply not possible (World Bank 2014a).⁵ As has been extensively documented in Haiti and globally, the poor constantly face high-stakes financial decisions, such as borrowing money at very high interest rates to survive liquidity short-

⁴ Easton and Fass also describe the monetary consumption benefits of education for poor families in urban Port-au-Prince in the late 1980s: many nonpublic schools offered students school meals and social insurance for their families, creating a large net benefit to poor families who contributed tuition payments. However, these benefits seem to have become much less common: by the 2002–3 school year, only 12 percent of nonpublic schools reported offering school meals.

⁵ It is important to note that in Haiti, as in many developing countries, public education was not entirely free: parents were required to make a small fee payment of 100 Haitian gourdes per school year until 2011, when this fee was abolished. However, the costs associated with attending—uniforms, books, transport, etc.—are still borne by parents.

ages (Easton and Fass 1989; Collins et al. 2009). In these contexts, investing in education could be extremely costly or nearly impossible for many families. In addition to the direct monetary costs of schooling, opportunity costs of losing out on children's labor (in the household or in the marketplace) and distance to schools (particularly for rural families) also serve to reduce effective demand.

At the time when the intervention in Haiti was designed, a USAID-financed initiative had just issued a report synthesizing these perspectives and identifying the primary determinants of household demand for education throughout the developing world (EQUIP 2007). On the basis of more than 120 separate studies, mainly in low-income countries, the authors highlighted four major factors driving or repressing demand: expected monetary and non-monetary returns to education; direct costs in the form of tuition and fees, uniforms, and materials; opportunity costs such as lost child labor; and distance to schools (plus related safety concerns), which impose additional direct or indirect costs on households in the form of transport expenses, time spent by adults taking children to school, or psychological costs of worrying about children. Although there was much variation due to contextual factors such as local market demand for child labor, the main conclusion was that household demand for education is highly sensitive to cost—and, as expected, poorer households tended to be more sensitive in this regard than more prosperous ones.

Beyond household calculations of costs and benefits, there is also a strong economic rationale for public financing of primary education. Much recent economic research documents how individual acquisition of education benefits society by making individuals more engaged and responsible citizens and better parents, as well as by potentially increasing the overall level of productivity and growth in the economy (Moretti 2006; Oreopoulos and Salvanes 2011). To the extent that individuals do not take these benefits into account when making investment decisions, government funding can increase the efficiency of investment in education.

In order to exploit what was perceived order to meet the high latent demand for primary education among the poor, the government of Haiti launched a new program aimed at increasing access by lowering the costs of attending school, with financial and technical support from the World Bank and the Caribbean Development Bank (CDB).⁶ The program, called the Programme de Subvention (and referred to as the “tuition waiver program,” or TWP, in English), provides an annual per-student payment to participating nonpublic schools that agree not to charge any form of tuition fees to students. The TWP was first rolled out in the Nippes and Artibonite Departments

⁶ The program would later be financed by other donors as well, including the Canadian International Development Agency (CIDA), the Global Partnership for Education (GPE), and the Inter-American Development Bank (IDB).

in 2007.⁷ Its primary objective was to reduce the direct costs that families experienced in sending children to school and thereby increase both enrollments and average duration of schooling.⁸

The theory of change behind the program is that if the direct costs of education are substantially reduced in nonpublic schools, this releases a binding demand-side constraint, allowing more parents to send their children to school and enable them to stay there. The approach of focusing on nonpublic schools was taken because of the public sector's limited size, and in order to exploit what was perceived to be existing excess capacity in nonpublic schools because parents could not afford to keep their children in the nonpublic institutions through the full 6-year primary cycle.⁹ In addition, in areas where schools were already full, it was expected that the tuition waiver incentive would trigger a supply response by private actors, encouraging them to start or expand nonpublic schools.

This theory of change was confirmed in the results of focus groups conducted by NGOs contracted to conduct independent program verification with parents of students attending treatment schools. As part of annual verification, the NGOs randomly selected five to 10 participating schools per department in which to organize focus groups with parents of students attending participating schools.¹⁰ These parents reported that the TWP had significantly reduced the financial burden of educating their children and helped alleviate their concerns about how to keep their children in school (FONHEP 2013; FPN 2013). Many parents reported episodes prior to the TWP of their children having been sent home from school for days, weeks, or even months because of lack of payment and the stress and unhappiness such episodes had caused their entire families. Importantly, while parents consistently expressed the importance of the program to their and their children's well-being, they also reported continued financial pressures from the costs of schooling that they continue to bear, including uniforms and transport.

Participation of schools in the TWP is subject to several conditions, aimed at preventing opportunistic efforts to profit from the program, improving the learning environment, and improving grade-for-age accordance. First, in order to qualify, schools need to have a permit to operate from the Ministry of Education. Second, only children entering grade 1 for the first time, be-

⁷ Departments are the first-level administrative division in Haiti. The country is divided into 10 departments.

⁸ The TWP is part of the *Stratégie Nationale d'Action/Education pour Tous* (SNA-EPT).

⁹ Observation of dozens of nonpublic primary schools in Haiti by the authors finds that the first three primary grades usually have two to three times as many students as the last three, because of high rates of repetition and dropout in the early grades.

¹⁰ In general, NGOs requested that school directors contact eight to 12 parents per school to participate in the focus groups. Given that directors had discretion about whom to invite, the focus groups may not capture the full range of parents' perspectives and experiences.

tween the ages of 6 and 8, are eligible for the subsidy.¹¹ Each entering grade 1 cohort is then supported for the subsequent years through grade 6, subject to compliance with the program rules. Third, in addition to not charging parents any fees, schools are required to provide students with at least three school textbooks. Finally, there is a limit of 45 children per class and a maximum of two classes per grade that can benefit from the program per school.

Compliance with these conditions has been verified annually by independent organizations hired by the ministry to carry out technical audits. All participating schools are subject to the technical audit. This has revealed that, in practice, the program has been implemented fairly faithfully according to its design, though not without its problems. As with nearly all development projects in Haiti, implementation suffered from a number of delays, which often resulted in schools receiving their payments late in the school year. While the majority of schools have largely complied, the technical audits find that most fail to fully comply with at least one of the conditions. Most common of these is the failure to provide students with the requisite number of books (more than a quarter of participating schools in the Department of Artibonite in the 2014–15 school year), although cases have also been documented in which parents have been charged fees (7 percent of schools in Artibonite in 2014–15) and the 45 student-teacher ratio limit has not been respected (9 percent in Artibonite in 2014–15). However, the ministry has taken little to no action to enforce the conditions (either by encouraging compliance or by sanctioning noncompliance).¹²

The amount of the subsidy was set at US\$90 per student in 2007, well above estimated average school tuition to account for the extra costs associated with providing textbooks, limiting class size, and improving school inputs. Covering these indirect costs was deemed of paramount importance to ensure a basic level of quality in schools.¹³ The subsidy amount has not changed since.¹⁴ Funds are managed by school management committees (SMCs), constituted by the school director, the president of the parents' committee, and representatives of teachers. Funds are transferred to dedicated bank accounts opened in the name of the SMCs, with signing authority given to both the school director and the president of the SMC. The money can

¹¹ This is a potentially binding constraint in many cases, as the average age of a new first grader is nearly 8 years old (World Bank 2014a).

¹² For more updated information on program performance, see the publicly available implementation status reports available through the World Bank website (<http://www.worldbank.org/projects/P124134/education-all-project-support-second-phase-education-all-program?lang=en>).

¹³ While annual data on school tuition are not available, as shown in tables 1 and 2 below, tuition was approximately \$20 in 2003 and rose quickly over time to reach approximately \$75 in 2012. Data on textbook prices from 2007 are not available; a back-of-the-envelope calculation using average prices in 2016 discounted for inflation suggests that in 2007, primary school textbooks cost roughly \$2 per book, implying that the subsidy amount would have easily covered the purchase of the three required books.

¹⁴ As mentioned in n. 13, rapidly rising tuition fees have eroded the value of the subsidy (which is paid in the Haitian gourdes equivalent of US\$90).

be used for any of 10 purposes outlined in the operational manual, including paying teacher salaries, small rehabilitation projects, and school feeding programs.

In order to target poor communities, the program relies on self-selection into participation by schools serving poor families. The level of the subsidy at \$90 per student is well below the level of tuition charged by those schools serving children in upper-income quintiles, and they therefore self-select out of the program. On the other hand, many schools serving poor children stood to benefit from participation, since they could earn more per child through the subsidy than by continuing to charge low levels of tuition. With these additional resources, it was expected that schools would invest in improvements to benefit students, such as upgrading poor infrastructure, hiring more effective teachers, and providing more pedagogical materials. However, participating schools did not receive any specific support from the Ministry of Education to help guide their investments, and other than meeting the program conditionalities and spending within the outlined purposes, schools were effectively free to spend the subsidies received as they saw fit.

As evidenced from the annual reports compiled by the NGOs contracted to conduct independent verification, the expenditure categories were respected by participating schools, with the largest amount allocated to teacher salaries. Nearly all participating schools spend part of the subsidy on textbooks for students, and nearly 90 percent spend resources on the infrastructure and materials side, namely, making small upgrades to classrooms, purchasing school furniture, improving roofs, and fixing doors and windows (FONHEP 2013).

Measuring Results

Using national school census data, we provide answers, albeit partial, to two critical questions regarding the program's impacts. First, did the program meet its primary objective of increasing poor children's access to schooling? While a direct answer to this question is not possible because of the lack of detailed enrollment data, we examine the size of student populations, as well as rates of progress through school, as proxy indicators for the impact of reducing financial barriers on access. Second, did program participation, and the extra resources it entailed, improve learning conditions within schools? To address this question, we utilize available measures of various school-level characteristics related to teachers, materials, and infrastructure as indicators of the conditions students experience within schools.¹⁵

¹⁵ Certainly, the ultimate objective of providing access to school is to increase students' learning. Unfortunately, data on student learning are not currently available but may be the subject of future research.

Initial Randomization, Identifying Schools across Years, and School Survival

In the school year 2008–9, because of a limited amount of program financing, the Ministry of Education and the World Bank agreed that only approximately 100 schools from each of the five newly participating departments in the country would be randomly selected among the 1,034 qualifying schools that applied across all five departments (Centre, Grand-Anse, Nord-Est, Nord-Ouest, and Nord). A total of 547 schools were selected to participate: 110 in Centre, 115 in Grand-Anse, 112 in Nord-Est, 111 in Nord-Ouest, and 99 in Nord. The random selection of schools was carried out in a meeting in Port-au-Prince attended by the nine-member steering committee of the tuition waiver program.¹⁶ In the meeting, the director of the Ministry of Education's Department for Private Education and Partnership (DAEPP) used Microsoft Excel to assign a random number to each qualifying applicant school. Schools were then sorted by their random number, with a separate list for each department, and those with the lowest random numbers were selected to participate (up to the allocated total for each department, which had been assigned on the basis of the number of schools that applied). The randomization was conducted so that at least one school from each commune of each of the departments was represented in the sample, in order to favor rural areas and maximize the geographical distribution of the program.¹⁷ Education indicators have historically been worse in rural areas; for example, as of 2012, children living in rural areas are two percentage points less likely to be in school and 19 percentage points more likely to be over-age for their grade compared to children living in urban areas (World Bank 2014a). All applicant schools were then notified through letters from the Ministry of Education on whether or not they had been selected to participate, and all selected schools in fact chose to participate.

Following a large economics literature, we exploit the “natural experiment” created by the random selection of schools into the TWP to estimate the program's impacts. This literature takes advantage of naturally occurring random variation—ranging from acute weather events to policy decisions—to estimate the causal impacts of a range of factors on outcomes of interest. While this approach is attractive given its low cost and ease relative to a purposeful randomized experiment, major challenges with the interpretation of such estimates lie in the validity of the assumption of randomness and in the generalizability of the results (Rosenzweig and Wolpin 2000). In the case

¹⁶ The steering committee comprised representatives from the Ministry of Education (2), Ministry of Finance (2), nonpublic education sector (2), teachers union (2), and national association of parents (1).

¹⁷ In practice, this meant checking ex post that at least one school from each commune had been selected into participation, and in cases in which this was not true, to manually add the school with the lowest random number from each unrepresented commune and remove the school with the highest random number that had been selected into participation (assuming another school from the same commune was already included). The geographic divisions of Haiti are department, *arrondissement*, commune, and section *communale*. There are 140 communes in the country.

of the TWP, the selection procedure was purposefully random (and is assessed below). However, it is important to note that qualifying schools are not representative of the totality of schools, but rather represent those schools that meet the eligibility criterion of having a permit to operate from the Ministry of Education. In particular, as shown in table 1, qualifying schools that applied to the TWP charged slightly lower tuition, had larger student bodies, and were founded somewhat earlier than the average nonpublic school. At the same time, the schools are similar on student-teacher ratios and the number of shifts taught. On balance, while any results may not be applicable to all schools, they would likely be relevant to the broader population of nonpublic schools in Haiti.

The available waves of the national school census (2002–3 and 2011–12) provide information on school staffing and infrastructure, as well as the size of the student population.¹⁸ Because the schools were randomly selected when the program was expanded in the 2008–9 school year, estimates of the program's impacts on these measures can be made. As mentioned above, a direct measure of the program's results in terms of its primary objective—increasing the number of children in primary school—is not possible. Such a measure would require baseline and follow-up data on overall school enrollment in local communities where qualifying schools are located, and these data were not collected. In addition, the approach of selecting at least one school per commune to provide access to the program across a wider geographic area also ensures that program schools are not closely clustered, making it more difficult to measure effects on area enrollment rates.

In order to assess the success of randomization and so the extent to which causal impacts of the program on schools can be estimated, the 1,034 qualifying schools that applied were identified first in the 2011–12 school census and then matched back to their entries in the 2002–3 school census. Given the time lapse between the census waves and program rollout, the rapid turnover of nonpublic schools in Haiti, and the lack of census identifiers in the data collected from qualifying schools, only 64 percent of the schools chosen in 2008 were identified in the 2011–12 census.¹⁹ In the next step, 55 percent of these schools were then matched back to the 2002–3 census. In other words, 35 percent of the 1,034 qualifying schools are observed across the two census waves (table 2).

¹⁸ Perspectives on the quality of the school census data are provided below. The 2011–12 census was conducted less than 2 years after the January 20, 2010, earthquake. While the five departments studied here were not directly affected, population movement following the earthquake may have affected the schools in the study. This is particularly true in Centre, the closest of the five departments to Ouest.

¹⁹ Identification is done manually using commune, school name, and school address. Spelling variations and slight differences in information recorded in the TWP application data and across census years required that each school be manually identified in the 2011–12 census and then matched back to the 2002–3 census.

TABLE 1
SCHOOL SAMPLE COMPARED TO ALL NONPUBLIC SCHOOLS

	Mean Values Observed in 2003	
	Qualifying Applicant Schools	Nonpublic School Average
School founding year	1984	1988
Average annual tuition (Haitian gourdes)	581	777
Number of shifts taught	1.0	1.0
Total number of students	191	124
Student-teacher ratio	30	30
Observations	354	13,983

SOURCE.—World Bank staff estimates using Haiti National School Census, 2002–3.

Treatment and control schools were equally likely to be identified in the 2002–3 school census and had similar characteristics, suggesting that randomization was successful. Since baseline data were not collected in 2008, we use the 2002–3 school census data as a proxy for baseline, or pretreatment, data. Following randomization in 2008, an initial test for balance was conducted in which 64 percent of treatment and 68 percent of control schools were successfully identified in the 2002–3 school census. These results, presented in the appendix, also demonstrated that treatment and control groups were observationally equivalent at that point. However, the fact that a large share of both treatment and control schools were not matched back to the census data is a cause for concern. There are three likely causes for the low matching rate: (a) the lack of consistent unique identifiers for schools; (b) the generally low quality of the census data; and (c) the fast-changing nature of the education sector, where nonpublic schools frequently change their names and/or locations. While data collection approaches have varied somewhat over time, in general, the school census in Haiti is carried out manually, with school directors completing paper questionnaires with the support of local inspectors, who then send these questionnaires to Port-au-Prince for processing and data entry. The manual nature of the process and the lack of consistent error-checking procedures are believed to substantially reduce the

TABLE 2
SCHOOL SAMPLE AND IDENTIFICATION RATES

	Number of Qualifying Applicant Schools in 2008		% Identified in 2011–12 School Census		% Matched to 2002–3 School Census	
	Treatment	Control	Treatment	Control	Treatment	Control
Nord	99	114	84	660	55	56
Nord-Est	107	139	76	28	56	33
Centre	110	59	65	53	44	74
Grand-Anse	113	58	80	55	49	44
Nord-Ouest	110	115	75	58	63	69
Total	539	485	76	50	53	57

overall quality of the data. In aggregate, these factors create a major challenge for consistent identification of schools. While this adds appreciable “noise” to the exercise, it is not likely to have introduced a major bias, since these factors are assumed to have applied to treatment and control schools equally.

Treatment schools appear more likely than control schools to have remained open through the 2011–12 school year. As shown in table 2, schools participating in the TWP were much more likely than control schools to be identified in the 2011–12 census: 408 of the 539 (76 percent) participating schools were identified, compared to 244 of the 485 (50 percent) control schools. However, we cannot say with certainty that the schools not identified in the census had closed down. As mentioned above, without unique identifiers, tracking schools over time becomes very difficult, and even small changes in name and street address can cause schools to fall out of the sample. There are two factors that may account for the different degrees of success in identifying treatment and control schools in the 2011–12 census. First, having received consistent financing for several years, and knowing that they could count on continued transfers through the TWP for the next few years, participating schools may have been more likely to remain open, in the same location and with the same “branding,” than control schools. Second, because treatment schools had more contact with the Ministry of Education through the TWP, they may have been more readily locatable when it came time to conduct the census. However, because the TWP and census are run by separate departments within the ministry and one has no implications for the other (e.g., no data from the census are used for the TWP), this seems unlikely to be an important factor.

Across both census waves, the low rates of school identification are likely driven by both random and nonrandom attrition, as well as sources of error in the data. As discussed above, the lack of consistent school identifiers and the poor quality of the school census data contributed to limiting the proportion of schools successfully matched, and these issues seem likely to have had similar effects on treatment and control schools. Treatment also affects attrition: there is a 26 percentage point difference in the identification rate between the treatment and control schools with the 2011–12 census. This differential survival may have implications for the measures of the program’s impact on school characteristics: as in all randomized experiments, attrition that is correlated with the treatment can bias impact estimates. In this context, survival depends in large part on a school’s ability to attract students and operate in a financially sustainable manner. If attriting schools are those that would have had the fewest students, then the impact estimates may be understated.

Using the 2002–3 school census, a test for balance between the treatment and control schools that are identified in both census waves also suggests that even after selection (only nonattriters are observed) the groups appear

to have been observationally equivalent prior to the program. Schools are compared across five key types of characteristics measured in the school census: scope (levels taught and number of shifts), infrastructure (building materials, water, latrines, etc.), physical materials (desks, chairs, blackboards, etc.), staffing, and students (number, gender, and repetition rates).²⁰ As shown in table 3, treatment schools were founded earlier than control schools, on average, and have slightly more students in grades 5–6 (the final two grades of primary school). Schools founded earlier may be more well regarded in the community, in part because they may have been more likely to have educated children's own parents. While this factor did not create a significant difference in enrollment prior to the TWP, it may have helped parents decide to send their children to treatment schools once the program started. Along all other dimensions, treatment and control schools observed in both waves of the school census (and therefore having survived from 2002–3 through 2011–12) were observationally equivalent in 2002–3.

Results: Changes in School Characteristics

To estimate the effects of the program on participating schools, we compare characteristics of treatment and control schools using the 2011–12 school census. Because qualifying applicant schools were observationally equivalent in the 2002–3 school year and were randomly selected into (treatment) and out of (control) program participation, any subsequent differences observed between average characteristics of treatment and control schools can be plausibly attributed to participation in the program. This causal impact of the program on school characteristics can be estimated from a set of simple ordinary least squares regressions of the following form:

$$Y_s = \alpha + \beta T + X_s + \epsilon_s,$$

where Y_s is the outcome of school s as measured in the 2011–12 census, T indicates assignment to the treatment group, and X_s indicates the commune in which school s is located, in order to account for the fact that the probability of selection into the program differed across communes (Duflo et al. 2006). Each regression measures the impact of T (program participation) on a different Y_s outcome indicator. Results are presented in table 4, in which each row corresponds to the estimate of β for each regression.²¹

²⁰ Following Kling et al. (2007), indices are constructed to measure infrastructure and physical materials (separately) in order to avoid spuriously significant correlations due to a high number of variables. Each index is constructed as the equally weighted average of z -scores of each included variable, where more beneficial outcomes have higher scores. The infrastructure index includes variables measuring the existence of a sports field, library, director's office, recreation area, kitchen, latrines, water, electricity, as well as the quality of roof, wall, and floor materials. The materials index includes variables measuring the number of desks, chairs, and blackboards.

²¹ Results run on the 652 schools matched only to the 2011–12 school census (rather than to both 2002–3 and 2011–12) are very similar to those presented in table 4.

INCREASING ACCESS BY WAIVING TUITION

TABLE 3
COMPARISON OF TREATMENT AND CONTROL GROUPS

	Mean Values Observed in 2003		
	Treatment	Control	<i>p</i> -Value
School founding year	1981	1986	.004**
Average annual tuition (Haitian gourdes)	611	562	.578
Scope:			
Number of school levels taught	2.1	2.1	.698
Number of shifts	1.0	1.1	.177
Infrastructure index	-.04	-.09	.147
Materials index	-.03	-.06	.445
School has a canteen (%)	17.9	18.2	.941
Staffing:			
Number of staff	6.35	5.93	.215
Share of male teachers	.69	.73	.154
Student-teacher ratio	35	37	.422
Students:			
Total number of students	198	186	.382
Grades 1–4: total number of students	152	148	.713
Grades 1–4: number of female students	71	73	.711
Grades 1–4: number of male students	81	75	.351
Grades 5–6: total number of students	47	38	.083*
Grades 5–6: number of female students	22	19	.156
Grades 5–6: number of male students	24	19	.059
% repeaters grades 1–6	.15	.14	.761
Observations	354		

SOURCE.—World Bank staff estimates using Haiti National School Census, 2002–3.

NOTE.—Following Kling et al. (2007), the infrastructure and materials indices are the equally weighted average *z*-scores of each component, where more beneficial outcomes have higher scores. The infrastructure index includes measures of a sports field, library, director's office, recreation area, kitchen, latrines, water, electricity, roof, wall, and floor materials. The materials index includes measures of desks, chairs, and blackboards.

* Significant at the 10% level.

** Significant at the 5% level.

The TWP permits a fairly wide range of potential uses of the subsidies, as discussed above, so directors of treatment schools could have invested in improving existing infrastructure, expansion, staffing, furnishings, learning materials, and so forth. Our results are limited, however, to the data available in the school census on the five areas of school characteristics measured in the census: scope (levels taught and number of shifts), infrastructure (building materials, water, latrines, etc.), physical materials (desks, chairs, blackboards, etc.), staffing, and students (number, gender, and repetition rates).

Across departments, treatment schools did not change the number of shifts offered, nor did they change the levels of school taught. The majority of treatment and control schools teach one shift per day and offer preschool and primary. The treatment does not appear to have had substantial effects on prompting improvements in infrastructure and physical materials: both coefficients are positive, but only the materials index is marginally significant. It is important to note that the average level of infrastructure and materials is low in both participating and control schools; for example, the majority lack sufficient chairs and desks for all students. The number of staff did increase by nearly one person, on average, in treatment schools.

TABLE 4
ESTIMATED IMPACTS OF TREATMENT OBSERVED IN 2011–12 SCHOOL YEAR

	$\hat{\beta}$	Standard Error
Scope:		
Number of school levels taught	-.01	.04
Number of shifts	-.01	.01
Infrastructure index	.05	.05
Materials index	.09*	.05
Staffing:		
Number of staff	.93**	.38
Share of male teachers	-.42	2.28
Students:		
Student-teacher ratio	8.3***	1.68
Total number of students	84***	14.53
Grades 1–4: total number of students	88***	11.66
Grades 1–4: number of female students	41***	6.14
Grades 1–4: number of male students	47***	5.92
Grades 5–6: total number of students	1	4.71
Grades 5–6: number of female students	0	2.74
Grades 5–6: number of male students	1	2.30
% repeaters grades 1–6	-.10***	.02
% overage grades 1–4	-.09***	.03
% girls overage grades 1–4	-.10***	.03
% boys overage grades 1–4	-.09***	.03
% overage grades 5–6	-.06*	.03
% girls overage grades 5–6	-.07**	.03
% boys overage grades 5–6	-.06*	.03
Observations	354	

SOURCE.—World Bank staff estimates using Haiti National School Census, 2011–12.

NOTE.—Following Kling et al. (2007), the infrastructure and materials indices are the equally weighted average z-scores of each component, where more beneficial outcomes have higher scores. The infrastructure index includes measures of a sports field, library, director's office, recreation area, kitchen, latrines, water, electricity, roof, wall, and floor materials. The materials index includes measures of desks, chairs, and blackboards.

* Significant at the 10% level.

** Significant at the 5% level.

*** Significant at the 1% level.

The number of students, male and female, in 2011–12 is substantially higher in treatment compared to control schools. Notably, this increase is limited to grades 1–4, which correspond to the four cohorts funded by the TWP from 2008 through 2011; no increase is observed in higher grades, which were not yet funded by the program, which had been running for only 4 years (fig. 1). Across departments, treatment schools have, on average, 88 more students in grades 1–4 compared to control schools. This large and significant increase indicates the strong demand from families for education at lower cost. However, we do not know what share of the additional students came from other schools (including control schools) and what share had not been in school previously. In other words, a simple comparison of the student population between treatment and control schools as an estimate of the program's impact on the number of children in school could be misleading, as the stable unit treatment value assumption is violated: the participation of some schools in the TWP almost certainly affects the student population at control schools (Angrist et al. 1996).

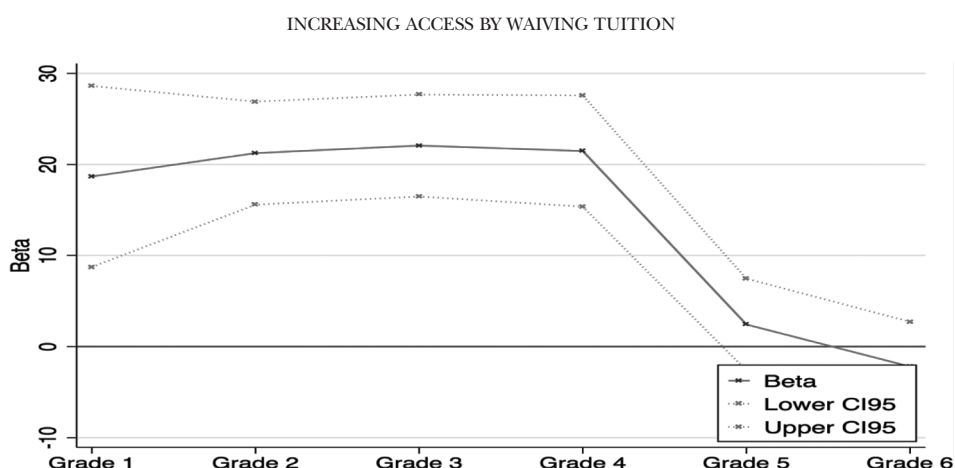


FIG. 1.—Impact of TWP on class size in grades 1–6. A color version of this figure is available online.

Figure 2 shows that between 2002–3 and 2011–12, the average number of students grew substantially in treatment schools while declining slightly in control schools across departments. Overall, the growth in treatment school student populations is larger than in control schools. Treatment schools gained an average of 78 students, while control schools lost 16 students. However, a share of the control schools not identified in the 2011–12 census may have closed down as a result of competition from treatment schools. Because treatment schools were geographically dispersed by design (with one per commune), 95 percent of control schools were located in the same commune as at least one treatment school. Communes (the second-largest administrative level) are fairly large, and even the least populated communes have over 12 primary schools. Therefore, it is not possible to conclude to what

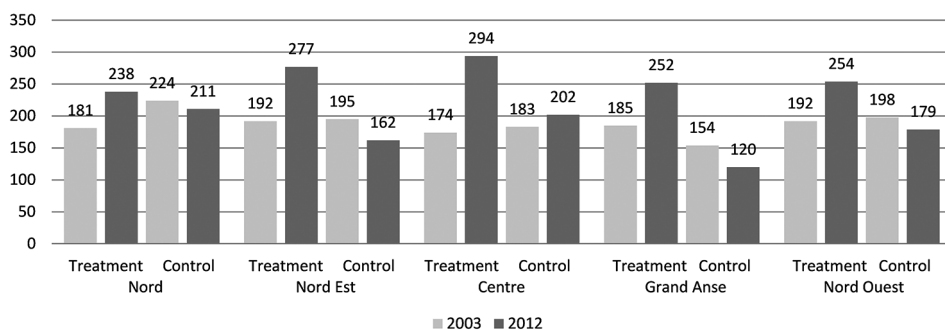


FIG. 2.—Total number of students (grades 1–6) in treatment and control schools (2002–3 and 2011–12). A color version of this figure is available online.

degree the program had an effect on overall enrollment in the communities affected.²² It is worth noting, however, that at the national level, the net enrollment rate at the primary level increased from approximately 50–60 percent in the early 2000s before the program to something like 70–80 percent in 2012 (based on data from the 2001 ECVH [Enquete sur les Conditions de Vie en Haiti], the 2005 and 2012 Demographic Health Survey [DHS], and the 2012 Enquete sur les Conditions de Vie des Ménages Après le Séisme [ECVMAS]).²³

Encouragingly, the percentage of primary students 2 or more years over-age for their grade fell by 10 percentage points in the treatment group. This is driven in part by lower repetition rates across grades, and these effects are similar for boys and girls. These impacts suggest that the reduced financial burden may allow students to participate in school more consistently. In addition, since the program funds only students who are within 2 years of the prescribed age for grade, schools also have an incentive to ensure that children do not repeat a grade multiple times. Interestingly, the effects are also seen in grades 5 and 6, not yet funded by the program, suggesting a potential spillover effect on nonsubsidized cohorts. Specifically, families may be better able to finance the education of older siblings when younger siblings are covered by the program, allowing them to stay in school more consistently and not have to repeat grades.²⁴

Finally, given the large increases in student numbers, the student-teacher ratio increased by eight on average; the average ratios remain under 40 in treatment schools across four of the five departments, well below the limit of 45 students per class mandated by the program (fig. 3). This increase in ratios may be a cause for concern, as the results of well-identified research, using natural experiments or randomization, suggest that class size differences of this magnitude can affect student learning and long-term outcomes, though the issue continues to be debated.²⁵

Conclusions

This article has sought to evaluate an important program in Haiti designed to increase primary school access in the context of low state capacity

²² In order to assess the impact of treatment schools on control schools' survival, more geographically disaggregated data at the time of program randomization would be needed.

²³ The 2001 ECVH measured net primary enrollment at 60 percent, the 2005 DHS at 50 percent, the 2012 DHS at 77 percent, and the 2012 ECVMAS at 72 percent (IHESI 2003; Cayemittes et al. 2007, 2013; World Bank 2014a). Differences in these measures arise both because they are sample surveys and because each was conducted in different months and years.

²⁴ Another potential cause for impacts on grades 5 and 6 repetition rates and share of students over-age is that schools utilized the resources of the program to improve the learning environment across grades, such that student learning and grade passing rates increased. However, given the lack of effects on measures of the learning environment, this possibility seems less likely.

²⁵ See Angrist and Lavy (1999); Krueger (1999); Hoxby (2000); Chingos (2010). Also see Duflo et al. (2015); Walters (2015).

INCREASING ACCESS BY WAIVING TUITION

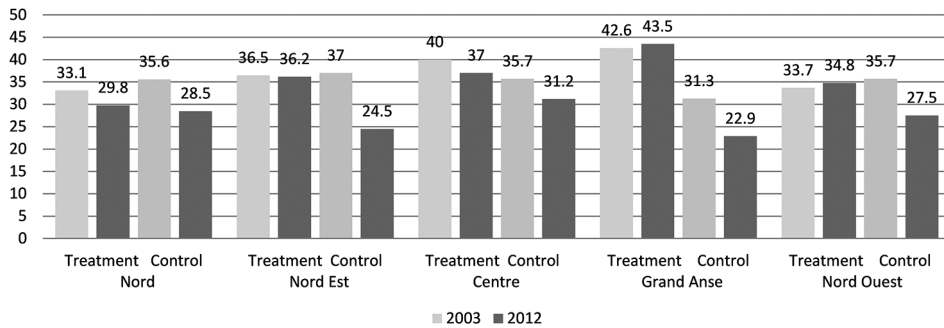


FIG. 3.—Average student-teacher ratios (grades 1–6) in treatment and control schools (2002–3 and 2011–12). A color version of this figure is available online.

by provision of tuition waivers that reduce the direct costs to the families of poor children. Our purpose is both to inform future policy development and practice in Haiti and to share lessons with those countries that still have large numbers of children outside the education system.

Prior to considering the results, the process of the study itself highlights several important lessons regarding the importance of collecting and using good-quality data for program evaluation, as well as policy making in the education sector as a whole. When the TWP started in 2007, 5 years had already passed since the last school census and the last population census. Since no additional data were collected for the program, government officials and program managers had limited reliable information on which to base decisions for allocating resources geographically. This constraint may have affected the program’s effectiveness. For example, as the program’s primary stated objective was to increase access, a data-driven approach to targeting the TWP to communes with the highest shares of out-of-school children could have increased its overall impact. In addition, the many limitations of the school census data, discussed above, coupled with the fact that they are not made publicly available, means that they are rarely used by researchers, constraining the production of knowledge on the education sector in Haiti. Without both reasonably good-quality data and useful research on key aspects of the education sector, policy makers and other actors have often been left to “fly blind” in their decision making, likely to the particular detriment of poor families. In the post-earthquake period of the last 5 years, increased attention has been given to data in many sectors including education. Three nationally representative household surveys have been conducted, a new population census is being planned, and several international organizations are supporting the Ministry of Education in particular to strengthen its data systems at all levels. However, continued financing and technical work from the government

and its development partners will be needed for many years to come in order to achieve sustained improvements.

Despite the challenges regarding the data, we are able to conclude that a school's participation in the TWP results in more students being enrolled, more staff, and higher student-teacher ratios. The program also reduces grade repetition and the share of students who are overage. While the increase in students at treatment schools cannot be proven to represent an actual reduction in the number of children out of school, it does demonstrate strong demand from families for their children's schooling. This demand is also evident in the results of focus groups conducted by NGOs, as previously discussed in the second section.

The program therefore seems to be achieving its objective of reducing the financial barriers to primary education faced by families in Haiti and may be supporting higher enrollment rates. Despite the program's estimated effectiveness, it has faced many operational challenges. Improvements in these areas—including enforcement mechanisms to create accountability for schools that do not provide all the inputs required by the program—could further increase the effectiveness of the program.²⁶ At the same time, abolishing fees cannot address the full set of financial barriers that keep children out of school in Haiti. The costs of uniforms, books and other materials, and transport are substantial, as are the opportunity costs for many households, who rely on children for domestic work or other labor. Addressing these issues requires more comprehensive social programs, such as the cash transfer schemes that exist in many Latin American countries (UNDG 2010). Moreover, participation in the program does not appear to improve learning conditions within schools, as measured by the availability and quality of infrastructure and materials. Therefore, moving beyond providing access to a classroom to ensuring that students learn will require greater investment, in terms of the waiver value, conditionalities, support provided to schools, and accountability.

The cost-effectiveness and financial sustainability of the TWP are additional important aspects to consider. If the net increase in student enrollment observed between treatment and control schools in fact represents students who otherwise would not have been in school and if all other students do stay in school regardless of the program (admittedly, two strong assumptions), then \$100 in waiver value results in an additional 0.25 year of student participation—outcomes that are in line with those found for several other interventions such as merit scholarships in Kenya (Dhaliwal et al. 2012; Evans and Popova 2014).²⁷ However, this rough back-of-the-envelope calculation does

²⁶ See, e.g., a recent evaluation funded by the Inter-American Development Bank (Forstmann and Cuenin 2014).

²⁷ Treatment schools gained an average of 78 primary students while control schools lost an average of 16, for a net gain of 62 (assuming no schools closed because of competition from the program). The estimate provided in the text assumes that all primary students in a treatment school (an average of 270 during treatment) must be provided a waiver in order to observe the net gain.

not account for the program's administrative costs on one hand nor the impact of the program on temporary and permanent dropout on the other. At a broader level, there is strong economic motivation for public financing of primary education as a public good, and this approach (public financing of private providers) may be more cost-effective and quicker than expanding public supply in Haiti. Reliable data on the costs of educating primary students in the public sector are not available, but building new public schools, a priority after the earthquake, has moved slowly and at high cost, and many costly inefficiencies continue to exist in the public system (Haiti Ministry of Finance 2012).

A cautionary note is in order regarding the critical role of strong oversight mechanisms for achieving these positive results. In 2011, the Haitian government initiated a program called PSUGO (Programme de Scolarisation Gratuite, Obligatoire, et Universelle—Program for Free, Obligatory, and Universal Schooling) with a component modeled closely after the TWP. Under this component, thousands of nonpublic schools received subsidies financed by national tax revenues to educate children tuition free. However, PSUGO lacked the TWP's strong monitoring processes and suffered from numerous cases of fraud, undermining its credibility and contributing to the government's decision to phase it out starting in 2014.²⁸ This experience highlights the fact that governments' role in education in no way diminishes when financing and provision are separated—public management and oversight become perhaps all the more important.

Taken together, the results from Haiti have several potential implications for policy to increase access further afield. Many countries with high numbers of out-of-school children have conditions similar to Haiti: these states' ability to provide services is low (either because of weak capacity or because of a lack of authority over parts of their territory), and they face problems preceded by political instability, climate change, or both. For example, UNESCO estimates that half of those out of school are living in conflict-affected countries. In these situations, private actors are often crucial to service delivery, and the government's role may be limited to financing the service and establishing and enforcing the governing mechanisms. This role is not an easy one to fulfill effectively, particularly in low-capacity settings, but with sufficient attention and resources it can be done. The success of the Haitian TWP in the face of multiple implementation challenges adds credence to the notion that public financing of nonpublic provision of services is a viable and promising approach for reaching those children still outside of the system. This bodes well for getting more children into school, keeping them there, and getting them to completion. Such results are certainly not enough to transform education in countries like Haiti, given the broader systemic challenges that

²⁸ The primary driver for the government's decision was that the program had expanded too quickly and did not have sufficient resources to cover its commitments.

such initiatives cannot address, but they represent a necessary and encouraging start.

As private schools catering to the poor multiply quickly, research on their impacts is growing in response, but many questions remain to be answered. From 1990 to 2010, the share of children in low-income countries enrolled in private schools has doubled from 11 to 22 percent (World Bank 2014b). Across Africa and South Asia, where these trends have been strongest, evidence suggests that these schools are relatively low cost, provide an education at least on par with public schools, and can increase enrollment rates (Andrabi et al. 2015). Going forward, several questions remain for researchers, regarding both the role of the private sector in basic education and the best approaches to addressing the unfinished agenda of universal enrollment.

Regarding the role of the private sector, two key questions for future research stand out. First, given the current realities of rapid growth, how can governments effectively regulate and in some cases finance private education providers? There is a strong economic rationale for not “picking winners” by financing individual schools in communities or particular school chains, but rather making financing available to all private providers in a community and supporting information dissemination to enable parents to make their own choices (Das 2016). However, evidence for how this works in practice is needed to help guide policy making. Second, what effects does a growing private sector have on segregation and inequality in education and on the public education sector? Answers to this question are not straightforward and are likely to depend on many country-specific factors, but research to address them is very much needed to inform the broader policy discussion around what should be the role of the private sector in providing basic education (Dixon 2012; Aubry and Dorsi 2016).

Finally, much more systematic research is needed on effective approaches to bring the remaining out-of-school children into the education system. These children are often the most vulnerable in society and include children facing extreme poverty, disability, and conflict. If the world is to meet the new, more ambitious goal of universal, quality education for all children under the Sustainable Development Goals, learning how to reach and educate these children will be critical.

Appendix

Analysis of Tuition Waiver Program Assignment Randomization

The World Bank tuition waiver program expanded to five new departments in the 2008–9 school year: Centre, Grand-Anse, Nord-Est, Nord-Ouest, and Nord. Lists were created in each department, including schools that were interested in participation and met the criteria of being nonpublic and having some level of Ministry of Education recognition. Tuition waiver recipient schools were then randomly selected from each province’s list.

In order to confirm that randomization had resulted in groups of schools that were equal in expectation at baseline, we compared selected and nonselected schools in the 2002 national school census, which was the most recent year available. Given the time lapse between 2002 and the initial list development in 2008, the rapid turnover of private schools in Haiti, and the fact that the 2008 school lists did not include census identifiers, not all 2008 schools were present or identifiable in the 2002 census. Percentages of applicant schools identified in the 2002 census are as follows: 61 percent in Centre, 64 percent in Grand-Anse, 59 percent in Nord-Est, 78 percent in Nord-Ouest, and 64 percent in Nord. Overall, 64 percent of selected schools were identified, and 68 percent of nonselected schools were identified.

Next we compared the identified selected and nonselected schools within departments on 19 variables in the 2002 school census, using chi-square tests and *t*-tests as appropriate. Across the 95 individual statistical tests, just five within-department comparisons showed statistically significant differences (see the bold comparisons in table A1). The five differences were all on different variables. The total number of statistically significant differences represent approximately 5 percent of all tests conducted, which would be expected. Therefore, the within-department randomization appears to have been correctly done, producing groups of schools that were equivalent in 2002.

Future evaluation work comparing selected and nonselected schools will focus on school size, and it is therefore important to note that there were no statistically significant differences between selected and nonselected schools in the number of children enrolled in first grade, overall or by gender.

TABLE A1
COMPARING SELECTED SCHOOLS TO NONSELECTED APPLICANT SCHOOLS IN THE 2002 CENSUS

Variable	Nonselected Schools (Mean or Percentage)	Selected Schools (Mean or Percentage)	<i>p</i> -Value (χ^2 or <i>t</i> -Test)
Centre			
Electricity	85%	84%	.887
Toilet	90%	83%	.362
Courtyard	14%	21%	.411
Desks	2.00	2.29	.615
Desks—good condition	1.18	2.21	.057
Chairs	4.53	4.25	.660
Chairs—good condition	3.24	3.60	.580
Kitchen	70%	75%	.659
Library	90%	95%	.369
License	7%	3%	.396
Year founded	1982	1986	.265
Management committee	49%	63%	.142
Parent committee	72%	85%	.109
Any committee	74%	88%	.067
Urban	23%	18%	.541
Private	84%	87%	.676
1st-grade enrollment	48.49	44.10	.592
1st-grade boys	23.65	22.00	.692
1st-grade girls	24.84	22.10	.515

TABLE A1 (Continued)

Variable	Nonselected Schools (Mean or Percentage)	Selected Schools (Mean or Percentage)	<i>p</i> -Value (χ^2 or <i>t</i> -Test)
Grand-Anse			
Electricity	92%	89%	.637
Toilet	97%	96%	.791
Courtyard	33%	52%	.065
Desks	2.81	2.21	.288
Desks—good condition	2.35	1.39	.112
Chairs	5.42	4.53	.228
Chairs—good condition	3.77	3.61	.806
Kitchen	90%	88%	.817
Library	94%	90%	.449
License	6%	4%	.734
Year founded	1982	1988	.022
Management committee	64%	49%	.151
Parent committee	75%	70%	.576
Any committee	81%	78%	.766
Urban	31%	30%	.964
Private	97%	99%	.606
1st-grade enrollment	52.61	45.07	.381
1st-grade boys	27.69	24.12	.398
1st-grade girls	24.92	20.95	.380
Nord-Est			
Electricity	83%	82%	.946
Toilet	74%	88%	.031
Courtyard	14%	14%	.921
Desks	3.90	3.04	.207
Desks—good condition	2.82	2.56	.692
Chairs	5.58	5.43	.793
Chairs—good condition	4.50	4.79	.656
Kitchen	51	57%	.493
Library	83%	89%	.271
License	5.48	4.00	.672
Year founded	1986	1988	.241
Management committee	78%	85%	.253
Parent committee	92%	93%	.719
Any committee	92%	95%	.484
Urban	53%	59%	.521
Private	99%	100%	.309
1st-grade enrollment	47.77	45.16	.606
1st-grade boys	24.49	23.12	.618
1st-grade girls	23.27	22.04	.621
Nord-Ouest			
Electricity	84%	88%	.420
Toilet	87%	91%	.450
Courtyard	10%	14%	.371
Desks	3.05	3.19	.787
Desks—good condition	2.07	2.39	.467
Chairs	6.07	5.45	.191
Chairs—good condition	5.00	4.28	.145
Kitchen	59%	62%	.724
Library	79%	90%	.048
License	5%	10%	.273
Year founded	1982	1983	.407
Management committee	87%	89%	.691
Parent committee	83%	88%	.349
Any committee	93%	99%	.046

TABLE A1 (Continued)

Variable	Nonselected Schools (Mean or Percentage)	Selected Schools (Mean or Percentage)	<i>p</i> -Value (χ^2 or <i>t</i> -Test)
Nord-Ouest			
Urban	22%	30%	.242
Private	100%	100%	n/a
1st-grade enrollment	50.41	48.08	.613
1st-grade boys	25.69	25.11	.809
1st-grade girls	24.72	22.98	.450
Nord			
Electricity	67%	69%	.767
Toilet	68%	77%	.247
Courtyard	19	38%	.013
Desks	4.40	3.61	.230
Desks—good condition	3.34	2.90	.498
Chairs	6.19	5.89	.636
Chairs—good condition	5.40	4.86	.438
Kitchen	51%	49%	.855
Library	85%	87%	.709
License	10%	4%	.163
Year founded	1982	1983	.726
Management committee	70%	75%	.552
Parent committee	69%	77%	.326
Any committee	81%	88%	.346
Urban	46%	45%	.905
Private	99%	100%	.404
1st-grade enrollment	45.73	51.05	.478
1st-grade boys	24.62	25.96	.753
1st-grade girls	21.11	25.09	.276

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